

Future RAG systems will incorporate more sophisticated reasoning capabilities. Rather than simple retrieval-then-generate, they will engage in multi-step reasoning, retrieving information, forming hypotheses, retrieving additional information to test these hypotheses, and iteratively refining their understanding.

This evolution toward reasoning systems will require closer integration with symbolic AI approaches, combining neural retrieval and generation with explicit reasoning over structured knowledge. The result will be systems that don't just retrieve and parrot information but genuinely reason with it.

Personalization and Adaptation

RAG systems will become increasingly personalized, learning from interactions with individual users to better understand their knowledge, preferences, and communication style. This personalization will extend beyond surface-level adaptation to fundamentally shape what information is retrieved and how it's presented.

Privacy-preserving personalization techniques will be essential, allowing systems to learn from user interactions without compromising sensitive information. Federated learning and on-device personalization may play important roles.

Ethical and Governance Frameworks

As RAG systems become more capable and widely deployed, robust governance frameworks will be essential. These frameworks must address questions of

accountability when systems provide incorrect information, transparency about information sources, and mechanisms for correcting errors or removing inappropriate content.

Standards for evaluating RAG systems across different dimensions—accuracy, fairness, robustness, transparency—will emerge, enabling meaningful comparisons and certifications. Regulatory frameworks may establish requirements for certain high-stakes applications.

Conclusion

Retrieval-Augmented Generation represents a fundamental shift in how we build systems that interact with knowledge. By combining the flexibility and fluency of large language models with the precision and updatability of retrieval systems, RAG addresses many of the limitations that constrained earlier approaches.

The journey from simple to advanced RAG systems has been marked by continuous innovation—better retrieval techniques, more sophisticated integration with generation, and growing understanding of how to evaluate and optimize these systems. Yet the field remains in its early stages, with many challenges still to address and opportunities yet to explore.

For practitioners and researchers, RAG offers a rich landscape for innovation. The basic architecture admits countless variations and enhancements, each suited to different applications and constraints. As the field evolves, we can expect RAG systems to become more capable, more

reliable, and more seamlessly integrated into how we work with information.

The future of RAG is not just about better retrieval or better generation in isolation, but about creating systems that genuinely understand and work with knowledge in ways that augment human intelligence. This vision—of AI systems that can access, reason with, and communicate knowledge effectively—is what continues to drive progress in this exciting field.

Field	Available from Docling?
File name, file path	Yes (but you already have this — you passed it in)
Number of pages	Yes
Document size	Yes (but you get this from the filesystem, not Docling)

Challenge:

Mehrfach gedacht, besser entschieden: Prio-1-Projekte mit Self-Consistency Prompting

Organisationen bearbeiten parallel eine Vielzahl interner Projekte, die um Ressourcen, Aufmerksamkeit und Priorität konkurrieren. Oft ist unklar, welches Projekt die höchste Priorität haben sollte, da verschiedene Kriterien (z. B. Relevanz oder Nutzen) abgewogen werden müssen. GenKI kann diesen Entscheidungsprozess unterstützen, indem sie Abwägungen transparent macht und verschiedene Argumentationslinien durchspielt. Eine dafür geeignete Methode ist Self-Consistency Prompting, bei dem die GenKI mehrfach unabhängig argumentiert und dann die Ergebnisse miteinander verglichen werden, um konsistente Muster zu erkennen.

HINTERGRUND

AUFGABE

Erstellt mithilfe der GenKI eine priorisierte Liste interner Projekte. Ausgangspunkt ist eine bestehende oder selbst erstellte Liste laufender/geplanter interner Projekte. Zunächst formuliert einen Prompt, der die GenKI dazu anleitet, die Priorisierung mehrfach unabhängig vorzunehmen (in verschiedenen unabhängigen Chats). Die GenKI soll dabei jedes Mal begründen, welches Projekt sie als höchste Priorität einstuft und warum. Anschließend vergleicht die unterschiedlichen Ergebnisse, identifiziert wiederkehrende Muster und prüft, welches Projekt über die verschiedenen Durchläufe hinweg am häufigsten als Prio-1-Projekt erscheint. Abschließend reflektiert:

Inwiefern führt Self-Consistency Prompting zu besseren Priorisierungsentscheidungen?

Nutzt selbst erstellte Beispiele oder sucht euch aus dem Internet/Intranet passende heraus.

MATERIAL